

■ Derivative

f' represents the derivative of a function f of one argument.

`Derivative`[$n_1, n_2, \dots$][f] is the general form, representing a function obtained from f by differentiating n_1 times with respect to the first argument, n_2 times with respect to the second argument, and so on.

f' is equivalent to `Derivative`[1][f]. ■ f'' evaluates to `Derivative`[2][f]. ■ You can think of `Derivative` as a *functional operator* which acts on functions to give derivative functions. ■ `Derivative` is generated when you apply `D` to functions whose derivatives *Mathematica* does not know. ■ *Mathematica* attempts to convert `Derivative`[n][f] and so on to pure functions. Whenever `Derivative`[n][f] is generated, *Mathematica* rewrites it as `D[f[#]&, {#, n}]`. If *Mathematica* finds an explicit value for this derivative, it returns this value. Otherwise, it returns the original `Derivative` form. ■ Example: `Cos' → -Sin[#1] &`. ■ See page 416. ■ See also: `D`, `Dt`.